

Post-Newtonian self-gravity in $2 \rightarrow 2$ cloud transitions: the h_{0i} and h_{ij} contributions

Internal note (sg_vs_si project)

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Abstract

The self-gravity (SG) contribution to $2 \rightarrow 2$ level transitions such as $211 \times 211 \rightarrow 322 \times \text{BH}$ has so far been computed with the Newtonian (h_{00}) kernel only. This note works out the first post-Newtonian (1PN, relative $O(\alpha^2)$) corrections to that amplitude from the gravitomagnetic field h_{0i} and the spatial metric h_{ij} . For comparison it also evaluates the remaining pieces of the harmonic-gauge one-graviton exchange: the kinetic correction to T_{00} , retardation, and the nonlinear black-hole $\times$ cloud (G^2) terms. Each piece is evaluated for four channels using the same Kerr Green's function as the self-interaction (SI) rates. **Result:**

- The h_{0i} term is the largest 1PN correction. It is negative, $\delta_{\text{GM}} \simeq -1.1 \alpha^2$ for the 211/322 channels and $\simeq -0.33 \alpha^2$ for the 322/544 channels. It reduces Γ_{SG} by about 20% at $\alpha = 0.3$ for $211 \times 211 \rightarrow 322$.
- The h_{ij} term is about $5\times$ smaller and positive.
- The sum of all computed pieces moves the SG/SI crossover in f_a by $\lesssim 10\%$.

The corrections are of the same size as the effect of using Kerr instead of hydrogenic eigenfunctions in the Newtonian term. A gauge-consistent 1PN treatment is therefore needed for $\lesssim 20\%$ accuracy on Γ_{SG} at $\alpha \gtrsim 0.3$, but not for the qualitative SG-vs-SI picture.

1 One-graviton exchange at 1PN

Conventions: signature $(-, +, +, +)$, $c = \hbar = 1$, $G = 1/M_{\text{Pl}}^2$. Matter couples to $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ through $S_{\text{int}} = \frac{1}{2} \int h_{\mu\nu} T^{\mu\nu}$. In harmonic gauge, $\bar{h}_{\mu\nu} \equiv h_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} h$ obeys $\square \bar{h}_{\mu\nu} = -16\pi G T_{\mu\nu}$. Integrating out the potential graviton (on shell, the quadratic action is $-\frac{1}{2}$ of the coupling) gives

$$S_{\text{eff}} = G \int dt \int d^3x d^3y \frac{T_{\mu\nu}(\mathbf{x}) T^{\mu\nu}(\mathbf{y}) - \frac{1}{2} T(\mathbf{x}) T(\mathbf{y})}{|\mathbf{x} - \mathbf{y}|} + \text{retardation}. \quad (1)$$

Expanding in $v^2 \sim \alpha^2$ (so $T_{0i} \sim \rho v$, $T_{ij} \sim \rho v^2$):

$$L_{\text{eff}} = G \int \int \frac{1}{r} \left[\frac{1}{2} T_{00} T'_{00} + \frac{1}{2} (T_{00} T'_{kk} + T_{kk} T'_{00}) - 2 T_{0i} T'_{0i} \right] + O(v^4), \quad r = |\mathbf{x} - \mathbf{y}|. \quad (2)$$

The three structures in (2) map onto the metric components as follows, using $h_{00} = \frac{1}{2}(\bar{h}_{00} + \bar{h}_{kk})$, $h_{0i} = \bar{h}_{0i}$, $h_{ij} = \bar{h}_{ij} + \frac{1}{2} \delta_{ij}(\bar{h}_{00} - \bar{h}_{kk})$:

- $\frac{1}{2} T_{00} T'_{00}$: Newtonian h_{00} sourced by T_{00} , including the $O(v^2)$ part of T_{00} .

- $\frac{1}{2}(T_{00}T'_{kk} + T_{kk}T'_{00})$: half of it is h_{ij} . That is $h_{ij} = -2\Phi\delta_{ij}$, sourced by T_{00} , coupling to the cloud's stresses T^{ij} . The other, equal, half is h_{00} sourced by T_{kk} : pressure contributes to the active mass, $\rho \rightarrow \rho + 3p$. Only the sum is fixed in harmonic gauge; both halves give identical transition amplitudes.
- $-2T_{0i}T'_{0i}$: **gravitomagnetic** h_{0i} exchange between mass currents.

Bošković–Savić (v2, Eqs. 27–29) write $\nabla^2 h_{00} = -8\pi T^{00}$, $\nabla^2 h_{0i} = 16\pi T^{0i}$, $\nabla^2 h_{ij} = -8\pi\delta_{ij}T^{00}$. These are the leading sources, and they stop at the Newtonian term. At 1PN, h_{00} also receives the T_{kk} source above. Retardation, $1/(\mathbf{k}^2 - k_0^2) = 1/\mathbf{k}^2 + k_0^2/\mathbf{k}^4 + \dots$, replaces $1/r \rightarrow 1/r - \frac{1}{2}k_0^2 r$ in the $T_{00}T_{00}$ term.

Point-particle check. For $T^{00} = m(1 + \frac{1}{2}v^2)\delta^3$, $T^{0i} = mv^i\delta^3$, $T^{ij} = mv^iv^j\delta^3$, Eq. (2) plus retardation gives, for one pair,

$$\frac{Gm_1m_2}{r} \left[1 + \underbrace{\frac{1}{2}(v_1^2 + v_2^2)}_{T_{00}} + \underbrace{\frac{(v_1^2 + v_2^2)}{h_{ij} + h_{00}[T_{kk}]}}_{h_{ij}} \underbrace{\frac{-4\mathbf{v}_1 \cdot \mathbf{v}_2}{h_{0i}}}_{h_{0i}} + \underbrace{\frac{1}{2}(\mathbf{v}_1 \cdot \mathbf{v}_2 - (\mathbf{n} \cdot \mathbf{v}_1)(\mathbf{n} \cdot \mathbf{v}_2))}_{\text{retardation}} \right],$$

i.e. the Einstein–Infeld–Hoffmann coefficients $\frac{3}{2}(v_1^2 + v_2^2) - \frac{7}{2}\mathbf{v}_1 \cdot \mathbf{v}_2 - \frac{1}{2}(\mathbf{n} \cdot \mathbf{v}_1)(\mathbf{n} \cdot \mathbf{v}_2)$ (Landau–Lifshitz §106). The EIH Lagrangian also contains G^2 terms. Of these, the ones involving two cloud quanta and the hole are

$$L_{G^2} = -\frac{G^2 M \mu^2}{2} \iint |\psi(\mathbf{x})|^2 |\psi(\mathbf{y})|^2 \left[\frac{1}{r_{xy}} + \left(\frac{1}{r_x} + \frac{1}{r_y} \right) \frac{1}{|\mathbf{x} - \mathbf{y}|} \right], \quad (3)$$

which are of the same order, $GM/r \sim v^2$.

2 Stress tensor of the cloud

With $\phi = (\psi e^{-i\mu t} + \bar{\psi} e^{i\mu t})/\sqrt{2\mu}$ and keeping the non-oscillating parts, the leading terms are:

$$T_{00} = \mu |\psi|^2 + \frac{1}{2\mu} |\nabla \psi|^2 + \dots, \quad (4)$$

$$T_{0i} = -j_i, \quad \mathbf{j} = -\frac{i}{2} (\bar{\psi} \nabla \psi - \psi \nabla \bar{\psi}) \quad (\text{mass current}), \quad (5)$$

$$T_{kk} = \frac{1}{\mu} |\nabla \psi|^2 - \frac{3}{4\mu} \nabla^2 |\psi|^2. \quad (6)$$

The form of T_{kk} includes the $O(\Phi_{\text{BH}})$ metric factors. In flat space one would get an extra $3V|\psi|^2$. With the metric factors included, $\int T_{kk} = 2K$, matching $\sum mv^2$ for particles. For a transition, every bilinear is taken between an absorbed state A and an emitted state B : $\rho^{BA} = \bar{\psi}_B \psi_A$, $\mathbf{j}^{BA} = -\frac{i}{2} (\bar{\psi}_B \nabla \psi_A - \psi_A \nabla \bar{\psi}_B)$, $t^{BA} \equiv \mu T_{kk}^{BA} = \nabla \bar{\psi}_B \cdot \nabla \psi_A - \frac{3}{4} \nabla^2 (\bar{\psi}_B \psi_A)$.

3 Amplitudes for $1 + 1 \rightarrow 3 + 4$

Each term of $-L_{\text{eff}}$ gives an amplitude with the same combinatorial factor as the Newtonian and SI vertices. Mode 4 is absorbed at the horizon or escapes to infinity. The bilinears are (41) at $\mathbf{x}$

and (31) at $\mathbf{y}$:

$$A_N = -G\mu^2 \iint \frac{\rho^{41}(\mathbf{x})\rho^{31}(\mathbf{y})}{|\mathbf{x} - \mathbf{y}|}, \quad (7)$$

$$A_{h_{0i}} = +4G \iint \frac{\mathbf{j}^{41}(\mathbf{x}) \cdot \mathbf{j}^{31}(\mathbf{y})}{|\mathbf{x} - \mathbf{y}|}, \quad (8)$$

$$A_{h_{ij}} = A_{h_{00}[T_{kk}]} = -\frac{G}{2} \iint \frac{\rho^{41}t^{31'} + t^{41}\rho^{31'}}{|\mathbf{x} - \mathbf{y}|}, \quad (9)$$

$$A_{T_{00}} = -\frac{G}{2} \iint \frac{\rho^{41}(\nabla\bar{\psi}_3 \cdot \nabla\psi_1)' + (\nabla\bar{\psi}_4 \cdot \nabla\psi_1)\rho^{31'}}{|\mathbf{x} - \mathbf{y}|}, \quad (10)$$

$$A_{\text{ret}} = +\frac{G\mu^2 k_0^2}{2} \iint \rho^{41}\rho^{31'}|\mathbf{x} - \mathbf{y}|, \quad k_0 = \omega_1 - \omega_3, \quad (11)$$

$$A_{G^2} = +G^2 M\mu^2 \iint \rho^{41}\rho^{31'} \left[\frac{1}{r_x r_y} + \left(\frac{1}{r_x} + \frac{1}{r_y} \right) \frac{1}{|\mathbf{x} - \mathbf{y}|} \right]. \quad (12)$$

The Newtonian line is the kernel $K_{\text{SG}} = -G\mu^2/|\mathbf{x} - \mathbf{y}|$ used in the SG/SI comparison. We report $\delta_X \equiv A_X/A_N$, so $\Gamma_{\text{SG}} \rightarrow \Gamma_{\text{SG}}[1 + \sum_X \delta_X]^2$.

Sources for mode 4. Mode 4 is not normalisable: it is a Green's-function solution with ingoing or outgoing boundary conditions. Every derivative is therefore moved off $\bar{\psi}_4$ by parts, giving $A_X = \int d^3x \bar{\psi}_4 S_X$. With $\Phi \equiv \int \rho^{31}/|\mathbf{x} - \mathbf{y}|$ and $\mathbf{A} \equiv \int \mathbf{j}^{31}/|\mathbf{x} - \mathbf{y}|$:

$$S_N = -G\mu^2 \psi_1 \Phi, \quad (13)$$

$$S_{h_{0i}} = -4iG \mathbf{A} \cdot \nabla\psi_1 - 2iG \psi_1 \nabla \cdot \mathbf{A} = -4iG \mathbf{A} \cdot \nabla\psi_1 + 2G\mu k_0 \psi_1 \Phi, \quad (14)$$

$$2S_{h_{ij}} = G \left[-\psi_1 \int \frac{\nabla\bar{\psi}_3 \cdot \nabla\psi_1}{|\mathbf{x} - \mathbf{y}|} + \nabla\Phi \cdot \nabla\psi_1 + \Phi \nabla^2\psi_1 - 6\pi\psi_1^2\bar{\psi}_3 \right], \quad (15)$$

and similarly for the rest. The h_{0i} step uses current conservation for the transition current, $\nabla \cdot \mathbf{j}^{31} = i\mu k_0 \rho^{31}$. The h_{ij} step uses $\int \nabla^2 \rho/|\mathbf{x} - \mathbf{y}| = -4\pi\rho$.

The last term of $S_{h_{ij}}$ is a *local* quartic piece generated by gravity at 1PN. It has the same form as the axion contact interaction, with effective strength $1/(8f^2) \leftrightarrow 3\pi G$ for h_{ij} alone.

4 Physics of the two pieces

h_{0i} (gravitomagnetism). Co-moving mass currents repel: the h_{0i} term in the EIH Lagrangian is $-4Gm_1m_2 \mathbf{v}_1 \cdot \mathbf{v}_2/r$. For transitions out of the superradiant states, both transition currents $\mathbf{j}^{41}$ and $\mathbf{j}^{31}$ inherit the azimuthal flow of ψ_1 . They are therefore aligned in the same way as the transition densities. The gravitomagnetic amplitude then opposes the attractive Newtonian one, $\delta_{h_{0i}} < 0$, with size $\sim 4\langle v_1 v_3 \rangle$:

- $4(\alpha/2)(\alpha/3) \simeq 0.67\alpha^2$ for $211 \rightarrow 322$, against the computed $1.1\alpha^2$;
- $4(\alpha/3)(\alpha/5) \simeq 0.27\alpha^2$ for the $322/544$ channels, against the computed $0.32\alpha^2$.

The suppression for higher- n states is simply their smaller orbital velocity, $v \sim \alpha/n$.

h_{ij} (stresses). The Newtonian potential also curves space, $h_{ij} = -2\Phi\delta_{ij}$. This couples to the kinetic stress T^{ij} of the cloud, which acts like a pressure and adds to the active mass. It therefore strengthens the attraction, $\delta_{h_{ij}} > 0$. Its size is $\sim \frac{1}{2}\langle v^2 \rangle$, roughly a quarter of the h_{0i} term.

5 Everything else at the same order

The pieces of (1) are individually gauge dependent; only the full $O(\alpha^2)$ amplitude is physical. Besides h_{0i} and h_{ij} it contains:

1. the kinetic correction to T_{00} , whose local form and the $O(v^2)$ normalisation of ψ are convention dependent;
2. retardation, which is small because the energy carried by the graviton is only the level spacing $k_0 = \omega_1 - \omega_3$;
3. the G^2 black-hole $\times$ cloud terms (3), which are *not* small;
4. the coordinate mismatch: the eigenfunctions are in Boyer–Lindquist coordinates, the kernels in harmonic ones ($r_h \simeq r_{\text{BL}} - M$), an $O(M/r) \sim \alpha^2$ effect;
5. the $O(\alpha^2)$ corrections to the eigenfunctions themselves, which already enter the Newtonian term through the Kerr eigenfunctions;
6. frame-dragging (BH spin) terms at 1.5PN, $O(\tilde{a}\alpha^3)$;
7. the oscillating (2μ) parts of $T_{\mu\nu}$ for a real scalar, which radiate rather than mediate the potential exchange.

Items 1–3 are computed below as indications of size. Items 4–5 are represented by the Kerr-versus-hydrogenic difference of the Newtonian amplitude, shown for comparison.

6 Method and checks

- The corrections are evaluated with hydrogenic eigenfunctions. They are themselves $O(\alpha^2)$, so using Kerr eigenfunctions would change them only at $O(\alpha^4)$. Mode 4 uses the Kerr Green’s function of `gf_radial`, as in the SG/SI rate comparison, and is projected onto its leading multipole with the $\Sigma = r^2 + a^2 \cos^2 \theta$ weight.
- Potentials are built multipole by multipole on an (r, θ) grid: 3000 log-spaced radii and 40 Gauss–Legendre nodes. $\mathbf{A}$ is computed from the Cartesian spherical components $j_{\pm} = j_x \pm ij_y$ and j_z , each with a definite azimuthal number. The $|\mathbf{x} - \mathbf{y}|$ kernel uses $|\mathbf{x} - \mathbf{y}| = \sum_L P_L(\cos \gamma) [r_{<}^{L+2}/((2L+3)r_{>}^{L+1}) - r_{<}^L/((2L-1)r_{>}^{L-1})]$.
- **Integration by parts and continuity.** Replacing mode 4 by a bound state ($|320\rangle$), $\int \bar{\psi}_4 S_X$ was compared with the original double integrals, in which the derivatives act on ψ_4 . They agree to 6×10^{-7} for h_{0i} and 5×10^{-6} for the stress term, at $\alpha = 0.05$ and 0.2 .
- All δ_X scale as α^2 at small α , and their imaginary parts are $\lesssim 10^{-13}$.

Code: `src/ongoing_tests/sg_vs_si/sg_1pn.jl` (julia `sg_1pn.jl test` runs the self-test) and `plot_1pn.py`.

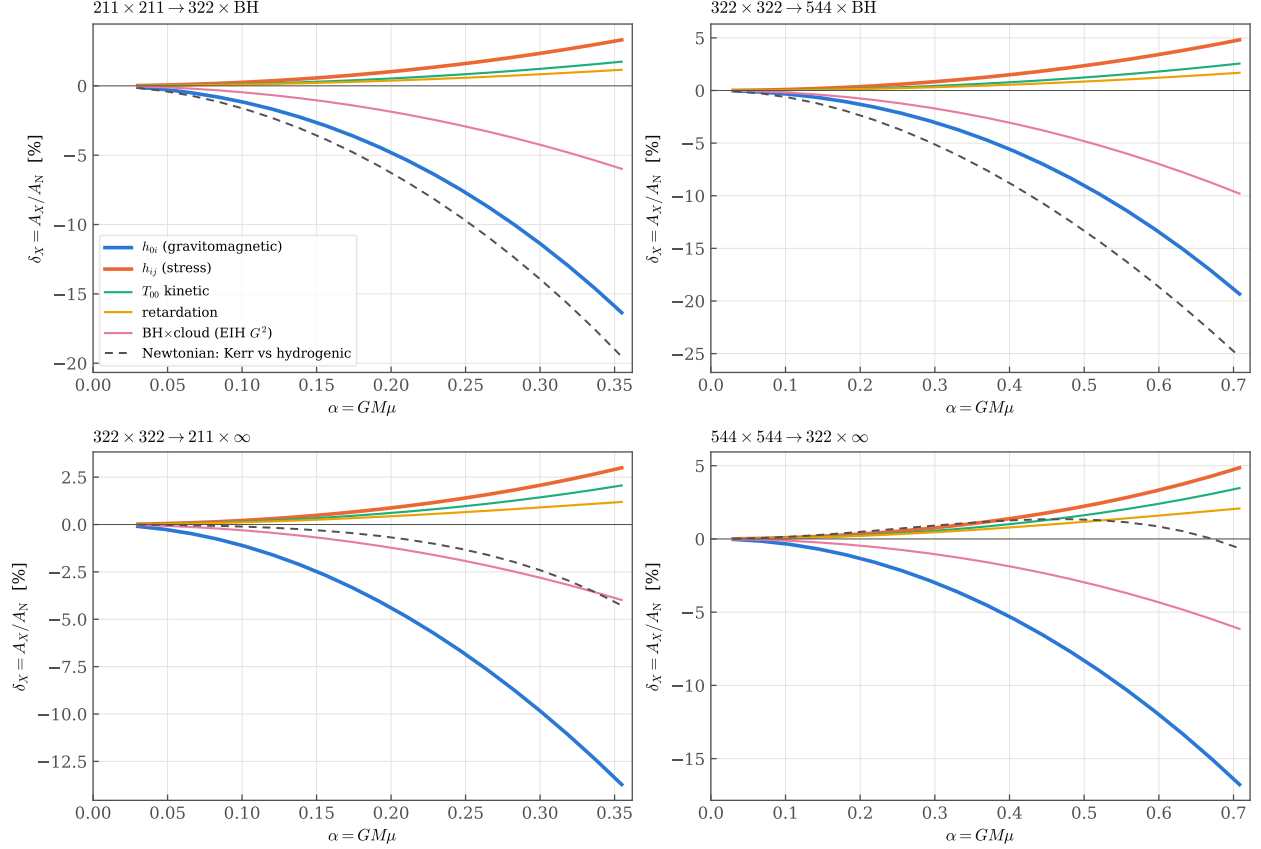


Figure 1: Fractional corrections $\delta_X = A_X/A_N$ to the Newtonian SG amplitude (Kerr BH, $\tilde{a} = 0.95$). Thick lines are h_{0i} and h_{ij} . The h_{ij} line is h_{ij} alone; the stress-sourced h_{00} adds an identical amount. The dashed line is the amplitude change in the *Newtonian* term from Kerr instead of hydrogenic eigenfunctions, $\sqrt{C_{\text{Kerr}}/C_{\text{hyd}}} - 1$, which represents the other $O(\alpha^2)$ effects (items 4–5).

7 Results

Conclusions.

1. h_{0i} is the largest 1PN correction to the SG-induced $2 \rightarrow 2$ amplitude. It is negative: $-1.1\alpha^2$ for $211 \times 211 \rightarrow 322 \times \text{BH}$ and $322 \times 322 \rightarrow 211 \times \infty$, and $-0.33\alpha^2$ for the $322/544$ channels. It lowers Γ_{SG} by $\simeq 20\%$ at $\alpha = 0.3$ in the former and $\simeq 6\%$ in the latter. At $\alpha \simeq 0.7$ the $322/544$ channels reach $\delta_{h_{0i}} \simeq -18\%$.
2. h_{ij} is $\sim 4\text{--}5\times$ smaller and positive: $+0.25\alpha^2$ and $+0.09\alpha^2$ respectively, or $+0.5\alpha^2$ and $+0.18\alpha^2$ if counted together with the stress-sourced h_{00} .
3. The G^2 black-hole $\times$ cloud terms are the second-largest piece and also negative. Retardation and the T_{00} kinetic term are at the 1% level for $\alpha \lesssim 0.3$.
4. The sum of the computed pieces is -9% , -2% , -6% and -1% in amplitude at $\alpha = 0.3$, for the four channels in table order. Because $\Gamma_{\text{SG}}/\Gamma_{\text{SI}} \propto C$, the crossover $f_a \propto C^{-1/4}$ moves up by only $|1 + \delta|^{-1/2} - 1 \lesssim 5\%$ at $\alpha = 0.3$, and $\lesssim 10\%$ over the full α range of each channel. The SG-vs-SI conclusions are therefore robust.

channel	h_{0i}	h_{ij}	T_{00}	ret.	G^2 BH $\times$ cloud	sum [†]
$211 \times 211 \rightarrow 322 \times \text{BH}$	-1.14	+0.25	+0.13	+0.10	-0.46	-0.87
$322 \times 322 \rightarrow 544 \times \text{BH}$	-0.32	+0.09	+0.05	+0.03	-0.19	-0.24
$322 \times 322 \rightarrow 211 \times \infty$	-1.10	+0.21	+0.15	+0.12	-0.30	-0.72
$544 \times 544 \rightarrow 322 \times \infty$	-0.33	+0.08	+0.06	+0.05	-0.11	-0.17

Table 1: Small- α coefficients δ_X/α^2 (at $\alpha = 0.03$). [†]Sum = $h_{0i} + 2h_{ij} + T_{00} + \text{ret} + G^2$, i.e. all computed harmonic-gauge pieces, with h_{ij} counted together with its equal $h_{00}[T_{kk}]$ partner.

channel	α	$\delta_{h_{0i}}$ [%]	$\delta_{h_{ij}}$ [%]	$\Delta\Gamma_{\text{SG}}$ from h_{0i} [%]	from h_{ij} [%]	sum [†] [%]
$211 \times 211 \rightarrow 322 \times \text{BH}$	0.30	-11.4	+2.4	-22	+5	-8.9
$322 \times 322 \rightarrow 544 \times \text{BH}$	0.29	-2.9	+0.8	-6	+2	-2.2
$322 \times 322 \rightarrow 211 \times \infty$	0.30	-9.9	+2.1	-19	+4	-6.2
$544 \times 544 \rightarrow 322 \times \infty$	0.29	-2.9	+0.7	-6	+1	-1.4

Table 2: Values at $\alpha \simeq 0.3$. $\Delta\Gamma_{\text{SG}} = |1 + \delta|^2 - 1$.

- These graviton-side corrections are comparable to the Kerr-versus-hydrogenic change of the Newtonian term. That change is -14% in amplitude at $\alpha = 0.3$ for $211 \rightarrow 322$, but $\lesssim +2\%$ for $544 \rightarrow 322$. A consistent 1PN calculation would combine both, with eigenfunctions and kernels in the same coordinates, and is required before quoting Γ_{SG} to better than $\sim 20\%$ at $\alpha \gtrsim 0.3$.
- The same kernels give the 1PN corrections to the self-gravity *level splittings* of Bošković–Savić (direct and exchange) by replacing mode 4 with a bound state. The h_{0i} exchange graph there is the missing diagram discussed earlier. This has not been evaluated here.

A Code fixes made alongside this note

- `spheroidals(1,m,a,erg)` passed $c = a\omega$, the massless convention of `SpinWeightedSpheroidalHarmonics`, so $c^2 = a^2\omega^2$. The massive angular equation needs $c^2 = a^2(\omega^2 - \mu^2)$. It is now `spheroidals(1,m,a,erg,alph)` with $c = a\sqrt{\omega^2 - \alpha^2}$ in *GM* units, in `solve_sr_rates.jl` and `MatterCouplings/sr_matter_ints.jl`. The call sites in `freq_shifts`, `s_rate_bnd` and `s_rate_inf`, which had passed ω in eV (i.e. $c \approx 0$), now pass dimensionless energies. Effect on `gf_radial`: $+0.08\%$ ($211 \times 211 \rightarrow 322$, $\alpha = 0.10$), $+2.1\%$ ($322 \times 322 \rightarrow 544$, $\alpha = 0.48$).
- In the ∞ branch of `gf_radial`, the flux radius was `itp_rrstar(itp_rrstar(r*))`, which maps $r_* \rightarrow r$ twice. It is now mapped once. Together with the above: $+0.7\%$ for $322 \times 322 \rightarrow 211 \times \infty$ at $\alpha = 0.17$.